Policy Paper

Closing the Last Mile in Deceased Organ Donation in India: Policy Frameworks, Grassroots Barriers and a District-Level Referral and Family Navigation Model

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ABSTRACT

India has an extensive legal and institutional framework for organ donation: the Transplantation of Human Organs and Tissues Act recognises brain-stem death, a national network of transplant organisations (NOTTO, ROTTOs and SOTTOs) coordinates allocation, and over five lakh citizens have registered Aadhaar-verified pledges since 2023. Yet deceased donation remains below one donor per million population, against about 48 per million in Spain. Of 20,138 transplants in 2025, only 3,526 used deceased-donor organs, while about 89,839 patients were waiting in March 2026. India therefore relies on living donors, about two-thirds of whom are women.

This paper argues that the shortfall is one of conversion, not willingness. Potential donors are lost between the ICU and the operating theatre: brain-stem death goes unrecognised or uncertified, hospitals without transplant programmes lack a referral pathway, families are approached late or without trained support, and medico-legal procedures delay retrieval. An original state-level analysis finds that five states with 27 per cent of India's population produced 82 per cent of deceased donors in 2023, while six large states with 42 per cent of the population produced under four per cent.

Three options are evaluated: expanding awareness and pledge campaigns, mandating identification and referral protocols, and a district-level referral and family navigation system. The paper recommends the third, incorporating the mandatory clinical triggers of the second. The proposed District Organ Donation Referral and Family Navigation Framework combines ICU identification triggers, a one-call referral line, shared rapid-response coordinators, multilingual family materials and stage-by-stage reporting with feedback to hospitals.

The framework should be piloted in ten districts across five states for 24 months, at an estimated INR 2.3–3.3 crore a year, with matched comparison districts and independent evaluation before national scale-up. Success should be judged by referrals, family approaches, consent rates and completed donations, not by pledges.

Keywords: gap in implementation, deceased organ donation, family consent, brain-stem death, donor referral systems, health policy in India

1. POLICY PROBLEM

1.1 The scale of unmet need

India performs more organ transplants than almost any other nation, but most rely on living donors. In 2024, 3,403 of 18,911 organ transplants used organs from deceased donors, and the deceased donation rate remained below one per million population (NOTTO, 2025a). A 2026 review ranked India first in the world for living donation and 68th for deceased donation, at 9.4 and 0.77 per million respectively (Kute et

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al., 2026). In 2025, total transplants rose to 20,138 and deceased donor transplants to 3,526 (NOTTO, 2026a), while about 89,839 patients were waiting for a major organ in March 2026 (Rajya Sabha, 2026).

The gap between need and supply is large. An estimated 175,000 kidney and 50,000 liver transplants are needed every year (NAMS, 2024), against 18,911 transplants of all organs in 2024 (NOTTO, 2025a). Meanwhile, about 1.73 lakh road traffic deaths occur annually, many involving the severe head injuries that can lead to brain-stem death, the most common route to deceased donation (NOTTO, 2025a).

1.2 A conversion problem, not a willingness problem

Deceased donation depends on a chain of events: a possible donor must be identified in the ICU, referred to the donation network, certified as brain-stem dead by an authorised board and physiologically maintained, and the family must be approached by trained staff so that it can make an informed decision. Organs must then be allocated, retrieved and transported. If any link fails, the process ends. India measures the start of this chain (pledges) and its end (transplants) but not the links in between, so most losses remain invisible.

The figures show the divide. More than five lakh Aadhaar-verified pledges were registered within three years of the national portal's launch, over 2.58 lakh of them in 2025 alone (Press Information Bureau [PIB], 2026; NOTTO, 2026a), yet India recorded only 1,235 deceased donors that year (NOTTO, 2026a). A pledge has no legal force at the time of death: the family must still consent, and a pledge matters only if the potential donor is identified and the family asked. The Observer Research Foundation concludes that India's constraint is not public unwillingness but low conversion, since many medically eligible deaths do not proceed to timely certification, retrieval and allocation (Athavle & Gopal, 2026).

This paper therefore focuses on the hospital stage of deceased donation: identification, referral, certification and family consent. Legal reform, allocation and circulatory death are treated briefly. Figure 1 outlines this pathway and the points where potential donors are lost. Only the last two stages, and the pledges outside the pathway, are counted nationally; identification and certification are counted partly and inconsistently, and family approach and consent not at all, so policymakers cannot see where most losses occur.

1.3 Who is affected

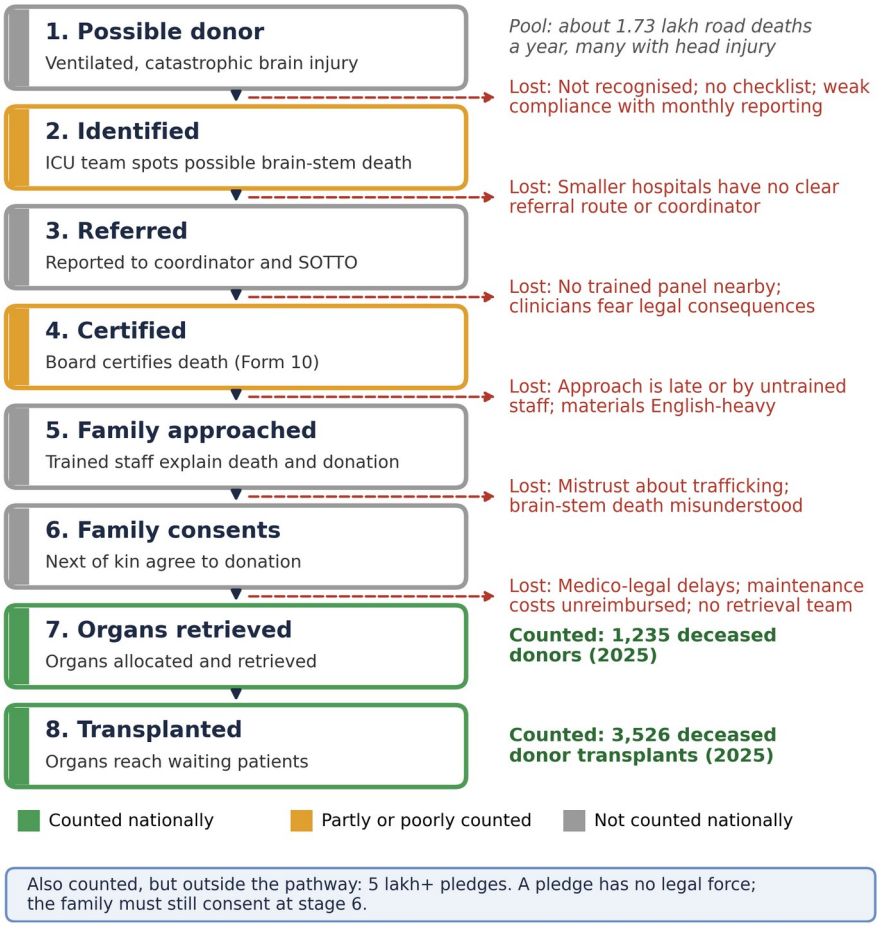
The gap harms patients with no willing family member; living donors, about 66 percent of whom are women (Kute et al., 2026); poorer and rural patients, since about 87 percent of kidney transplant centres are private and urban (Kute et al., 2026); families who face the donation question in distress and in need of guidance; and clinicians in non-transplant hospitals who may be unfamiliar with certification procedures and legal liability.

1.4 Why action is needed now

Two developments create an opening for reform. First, on 3 August 2026 the Ministry of Health issued NOTTO's National Guidelines on Brain Stem Death Certification, the first national framework for uniform certification, alongside e-Pratyaropan, a single digital platform for registration, allocation and monitoring (PIB, 2026). The guidelines propose nodal officers, audits and better family counselling but need a district-level delivery mechanism (NOTTO, 2026b, as reported in ETV Bharat, 2026), and e-Pratyaropan offers a channel for the reporting proposed here. Second, the current National Organ

Transplant Programme (NOTP) guidelines covered 2021–22 to 2025–26, so the design of its next phase is a natural point to introduce new elements.

Figure 1. The deceased donation pathway in India and where potential donors are lost



Source: Author's illustration based on THOTA procedures and the barriers documented in this paper (NAMS, 2024; Athavle & Gopal, 2026; NOTTO, 2025c). Counts from NOTTO Annual Report 2025–26 (PIB, 2026; NOTTO, 2026a).

2. POLICY CONTEXT

2.1 Legal framework

The Transplantation of Human Organs Act (THOA), 1994 responded to the kidney trade of the 1980s: it banned commercial dealing in organs, created committees to authorise living donation, and recognised brain-stem death as a legal basis for deceased donation. The 2011 amendment extended the Act to tissues (THOTA), mandated a national, regional and state transplant network, made transplant coordinators mandatory in registered hospitals, and required ICU doctors to ask whether a patient had pledged to donate and, if not, to inform the family of the option (Government of India, 1994; NOTTO, n.d.).

The procedure for brain-stem death certification was outlined under the Transplantation of Human Organs and Tissues Rules, 2014, which include a board of medical experts and most importantly, the nomination of a surgeon/physician and anaesthetist/intensivist where a neurologist/neurosurgeon is not available (Government of India, 2014; Srivastava et al., 2018).

2.2 Institutions and programmes

Transplant coordinators, retrieval and transplant infrastructure, organ transport, training and awareness are funded by the NOTP. The programme is coordinated by NOTTO, the apex body, which is backed by five Regional (ROTTO) and 26 State (SOTTO) organisations, and currently has 31 States and Union Territories participating with a total of approximately 1,200 registered hospitals (PIB, 2026). Hospitals can register as a transplant centre or a Non-Transplant Organ Retrieval Centre (NTORC) which can certify brain-stem death and retrieve organs but not transplant them.

2.3 Brain-stem Death: Recent Directions

A NOTTO letter of April 2024 asked registered hospitals to report brain-stem death cases to their SOTTO monthly and to display 'required request' boards in ICUs; a March 2025 notice reported continued non-compliance (NOTTO, 2025c). In November 2025 the Supreme Court directed States to act against hospitals that fail to report data to NOTTO (Indian Society of Organ Transplantation v. Union of India, 2025). The 2026 guidelines go further, proposing nodal intensivists, audits of missed opportunities, telemedicine support for smaller hospitals and medico-legal protection for certifying doctors (NOTTO, 2026b, as reported in ETV Bharat, 2026).

2.4 The policy–practice gap

Table 1 outlines the hospital and community level impacts of the national level weaknesses for each of the key provisions of the national framework.

3. EVIDENCE AND ANALYSIS

3.1 National trends

The number of transplants has been increasing over time but deceased donation has not increased as rapidly as living donation (Table 2). In 2023, for the first time in India's history, more than 1,000 deceased donors were recorded in a single year (Kute et al., 2026). However, by 2025, the deceased donor organ supply was still less than one in five of all transplants (PIB, 2026; NOTTO, 2026a).

3.2 Geographic concentration: an original analysis

National totals mask large differences between states. This paper calculated deceased donors per million population for each state with published figures. Donor counts for 2023 and 2024 come from NOTTO's state-wise data release for 2013–2024 (NOTTO, 2025b), cross-checked for the five leading states against Kute et al. (2026), and population denominators are official projections for 1 March of each year (National Commission on Population, 2019). Rates published elsewhere, including those often cited for Tamil Nadu, use different years and older population estimates and are not directly comparable. Table 3 and Figure 2 show the results; full calculations are in Supplementary Table S1 (Appendix B).

Table 1. National policy provisions and corresponding grassroots-level gaps

Policy provision	**Grassroots-level gap**
Citizens can register donation pledges online	Families may not know about the pledge, or may not realise that their consent is still required at the time of death
Hospitals must identify and report potential brain-stem death cases	Staff in smaller hospitals may lack training or a clear referral pathway; NOTTO reports continued non-compliance with monthly reporting
Brain-stem death is legally recognised under THOTA	Families may read ventilation and a beating heart as signs of life; other statutes do not define death in the same way, creating clinician hesitation
Registered hospitals must appoint transplant coordinators	Coordinator availability and training are uneven, especially outside large private hospitals, and coordinators are often overstretched
NOTTO, ROTTOs and SOTTOs coordinate donation and allocation	Citizens and smaller hospitals often do not know which organisation to contact, or when
National and regional awareness campaigns	Campaigns measure pledges and outreach rather than family conversations, referrals and completed donations
Official information and statutory forms are publicly available	Materials are technical, fragmented and English-heavy, and hard to use during an emergency
Allocation is governed by formal, rule-based systems	Mistrust about trafficking, favouritism and commercialisation continues to shape family decisions
Retrieval infrastructure exists in registered centres	District and rural hospitals are rarely registered as NTORCs and may be unable to identify, maintain or refer donors

Source: Author's analysis, drawing on NOTTO (2025a, 2025c), NAMS (2024) and Athavle and Gopal (2026).

Three findings stand out. First, in 2023 the five leading states (Telangana, Tamil Nadu, Karnataka, Maharashtra and Gujarat) held 27 percent of India's projected population but produced 82 percent of deceased donors. Second, six large states (Uttar Pradesh, West Bengal, Madhya Pradesh, Rajasthan, Odisha and Chhattisgarh), with about 42 percent of the population (577 million people), recorded only 41 donors, or 0.07 per million; Uttar Pradesh's rate was about 500 times lower than Telangana's. Third, had those six states matched Tamil Nadu's 2023 rate of 2.32 per million, they would have produced about 1,340 donors, more than the national total that year.

Table 2. Organ transplantation in India, 2023–2025

Indicator	2023	2024	2025
Total organ transplants	18,378	18,911	20,138
Transplants using deceased donor organs	2,935	3,403	3,526
Deceased donors	1,099	1,128	1,235
Deceased donation rate (per million population)	< 1	< 1	< 1

Sources: 2023 and 2024 from NOTTO Annual Report 2024–25 (NOTTO, 2025a), NOTTO (2025b) and Kute et al. (2026); 2025 from NOTTO Annual Report 2025–26 (NOTTO, 2026a; PIB, 2026).

Two checks suggest these results are robust. The concentration result holds in 2024, when the top five states had 27.4 percent of the population and 80.6 percent of donors (NOTTO, 2025b). The six-state result is also robust to error: even if their true donor counts were five times higher, their combined rate would be 0.36 per million, under half the national average. The analysis also shows what is at stake. Had every state matched Tamil Nadu's 2024 rate of 3.48 per million, India would have recorded about 4,860 deceased donors in 2024 instead of 1,128; at roughly three organs transplanted per donor, that implies about 11,000 more transplants a year from deaths that already occur (author's calculation; Supplementary Table S1). This is an upper-bound illustration: few states could quickly reach Tamil Nadu's rate, which is itself slightly inflated because donors are counted where donation takes place.

In 2024, much of the North-East (except Manipur), Uttarakhand, Jharkhand, Jammu and Kashmir, Ladakh and the Andaman and Nicobar Islands reported no deceased donation (NOTTO, 2025a). Because the law and brain-stem death criteria are the same across India, differences this large reflect state and hospital systems, not public attitudes alone.

The analysis counts donors where donation occurs, which inflates rates in referral hubs such as Delhi, and uses projections based on the 2011 Census; it shows association, not cause (Appendix B).

3.3 Where the pathway breaks

Identification and reporting

Although the law requires hospitals to identify potential brain-stem death cases in ICUs, NOTTO has repeatedly cited poor identification and certification as a key constraint, most recently in a March 2025 notice on continued non-compliance (NOTTO, 2025c).

Certification

Variations in clinical practice and lack of trained health workers are also barriers in the determination of brain-stem death and these points are highlighted in the 2026 guidelines (NOTTO, 2026b, as reported in ETV Bharat, 2026). The legal position is also unclear. The Registration of Births and Deaths Act, 1969 and the Bharatiya Nyaya Sanhita, 2023 do not include brain-stem death in their definition of death, as a

result of which some doctors are concerned about police inquiries after their certification (Athavle & Gopal, 2026).

Ability to refer for and retrieve

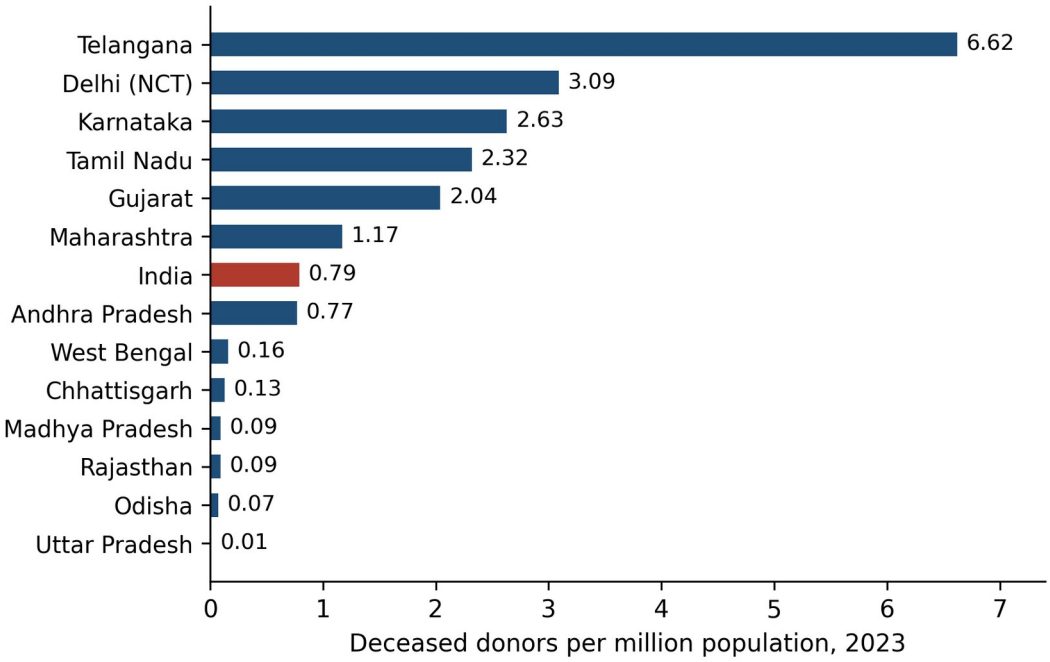
India's retrieval network is inverted: a functioning system would have many retrieval centres feeding fewer transplant hubs, but only about 140 NTORCs are registered against 618 transplant centres (National Academy of Medical Sciences [NAMS], 2024). The NTORC licence application has 49 fields (Athavle & Gopal, 2026). NOTP support for maintaining a deceased donor is INR 1 lakh per donor for only five donors a year, INR 5 lakh annually for the whole country, and only if the organ goes to a government hospital (NAMS, 2024; Government of India, 2021). A district hospital therefore has little reason to start a process it cannot fund or finish.

Table 3. Deceased donors per million population (pmp) by state, 2023 and 2024

State	Donors 2023	Population 2023 (million)	Donors pmp 2023	Donors 2024	Donors pmp 2024
Telangana	252	38.1	6.62	188	4.91
NCT of Delhi	66	21.4	3.09	n/a	n/a
Karnataka	178	67.7	2.63	162	2.38
Tamil Nadu	178	76.9	2.32	268	3.48
Gujarat	146	71.5	2.04	119	1.64
Maharashtra	148	126.4	1.17	172	1.35
Andhra Pradesh	41	53.2	0.77	n/a	n/a
West Bengal	16	99.1	0.16	n/a	n/a
Chhattisgarh	4	30.2	0.13	n/a	n/a
Madhya Pradesh	8	86.6	0.09	n/a	n/a
Rajasthan	7	81.0	0.09	n/a	n/a
Odisha	3	44.3	0.07	n/a	n/a
Uttar Pradesh	3	235.7	0.01	n/a	n/a
India	1,099	1,386.2	0.79	1,128	0.81

Source: Author's calculation (see Supplementary Table S1). Donors: Kute et al. (2026) and NOTTO (2025b). Population: National Commission on Population (2019), projected population on 1 March. n/a = state figure not separately available for 2024.

Figure 2. Deceased donors per million population by state, 2023



Source: Author's calculation from Table 3.

Family consent

Family consent is shaped by grief, trust and understanding. Disfigurement fears and concerns about funerals, as well as trafficking suspicions, all affect decisions. Fear of organs being misused or trafficked was the main reason for unwillingness to donate even in a survey of medical interns and residents in Gujarat (Tanna et al., 2023). Qualitative research with transplant coordinators points to the issue of being overworked, not having private counselling rooms and delays in the process as challenges in family discussion (Vincent et al., 2024).

Measurement

There is no routine national reporting of how many possible donors are referred, how many families are approached and consent, or why others decline. Without these data, losses cannot be located or hospitals held accountable.

3.4 Stakeholder perspectives

Each stakeholder group experiences a different part of the problem. ICU doctors and nurses fear certification errors and liability. Transplant coordinators report heavy workloads and weak institutional support (Vincent et al., 2024). Hospital administrators face complex registration and unreimbursed maintenance costs. Families need trustworthy information in their own language. Police are crucial in accident cases but rarely trained on brain-stem death.

3.5 Case studies and comparisons

Tamil Nadu

Tamil Nadu launched its deceased donor programme in 2008 through government orders that made brain-death declaration mandatory in registered hospitals. Government Order No. 289 of 2008 allowed any hospital with 25 beds, an operating theatre and an ICU to register as a non-transplant retrieval centre and certify brain death, with allocation coordinated centrally (MOHAN Foundation, n.d.). Its success is attributed to cooperation between government, private hospitals and NGOs and to transparent allocation (Abraham et al., 2012). More recently, TRANSTAN and Madras Medical College developed standard protocols for certification, donor maintenance and retrieval, and doctors from 25 government medical colleges were trained in 2024. That year the state recorded 268 deceased donors, up 51 percent on 2023, and government hospitals produced more donors than private ones for the first time (Natarajan & Ethirajan, 2026; NOTTO, 2025b). The lesson is that hospital readiness and a clear referral pathway, not new legislation, drove results.

Kerala

External doctors were added to panels of certifiers and apnoea tests were video recorded to increase public confidence (Mekkodathil et al., 2019).

Spain

Spain has the world's highest deceased donation rate, 49.4 per million in 2023 (The Lancet, 2024). The Organización Nacional de Trasplantes, created in 1989, built a network of hospital transplant coordinators, mainly intensive care doctors, who identify every donation opportunity and support families; this doubled Spain's deceased donation within a decade (Matesanz et al., 2017; La Moncloa, 2025).

3.6 Analysis

The evidence points to three conclusions. First, the binding constraint is institutional capacity at hospital and district level, not national law or public sympathy. Second, high performers combine mandatory identification, a clear referral route, trained family support and data on missed cases. Third, family consent depends on the institutional process: families are more likely to agree when it is timely, transparent and explained by trained staff in their own language. The policy response should therefore combine clinical accountability with family support, delivered close to hospitals.

4. POLICY OPTIONS

There are three options and each is examined below. Two additional reforms—the presumed consent and donation after circulatory death—were also considered but not shortlisted and are explained in section 4.4.

4.1 Option One: Proceed with and augment public awareness and pledge campaigns

Description. This option continues the current direction of national policy. It would increase the outreach of the year-long campaign of Jug Jug Jiyo Abhiyaan which began in August 2026, extend pledge drives in

colleges, workplaces and digital platforms, and continue to encourage people to participate in the Aadhaar-linked pledge portal (PIB, 2026).

Benefits. Campaigns are easy to run and politically visible, help normalise donation, and encourage families to discuss donation before a crisis (Athavle & Gopal, 2026). Engagement has clearly risen: more than five lakh Aadhaar-verified pledges were registered in three years, over 2.58 lakh of them in 2025 alone (PIB, 2026).

Costs. The proposed spending of the NOTP for 2021-22 to 2025-26 is INR 6 crore over five years (about INR 1.2 crore a year, or roughly INR 120 per portal pledge), for information, education and communication activities (NAMS, 2024).

Risks. The main risk is that awareness is not the binding constraint. Stated willingness is already high: in a 2018 Ipsos survey cited by the NAMS Task Force, 74 percent of Indian respondents said they would donate after death, against 72 percent in Spain (NAMS, 2024). Yet in 2025 India added about 200 pledges for every deceased donor, and donors rose only from 1,128 to 1,235 while pledges surged (NOTTO, 2025b; NOTTO, 2026a). Campaigns do not reach the hospital stages in Figure 1 where most losses occur, and a widening gap between pledges and donations may breed public cynicism.

Feasibility. High. The infrastructure already exists; it is effectiveness, not feasibility, that is in doubt.

4.2 Option Two: Increase the implementation of donor-identification and referral mandates in all hospitals with ICUs and trauma units

Description. This option would make identification and referral mandatory. All hospitals with an ICU or trauma unit would be required to screen ventilated patients with severe brain injury against a standard checklist, refer all suspected cases of brain-stem death to the SOTTO within a set period, and have to submit monthly reports, with penalties including the non-renewal of registration for failure to comply. The NAMS Task Force recommends a similar approach (NAMS, 2024).

Benefits. Mandatory identification addresses the first and largest point of loss. It creates institutional accountability, since hospitals would have to explain missed cases, and improves information on the size of the potential donor pool. There is precedent: Tamil Nadu's 2008 orders made brain-death declaration mandatory in registered hospitals, and the state has since led India's deceased donation (MOHAN Foundation, n.d.; Natarajan & Ethirajan, 2026).

Costs. The direct cost to government is moderate: the NOTP allocated INR 4 crore over five years for training and INR 3 crore for coordination with government medical colleges and well-performing private centres (NAMS, 2024), which could fund checklist rollout and monitoring. The larger costs fall on hospitals. At the NOTP rate of INR 35,000 per coordinator per month, employing a coordinator in every hospital would cost about INR 4.2 lakh per hospital per year (Government of India, 2021), rising with each hospital covered. Hospitals would also bear donor maintenance costs, for which central support is just INR 5 lakh a year nationally (NAMS, 2024).

Risks. Such mandates already exist: THOTA requires hospitals to identify potential donors, yet NOTTO issued a non-compliance notice in March 2025 (Government of India, 1994; NOTTO, 2025c). Penalties without support risk paper compliance, and hospitals without coordinators could refer but not follow through. In Kerala, deceased donors fell from 72 in 2015 to 8 in 2018, with the lack of brain-death

declaration in many hospitals cited as a main reason (Mekkodathil et al., 2019). The option also does nothing for family mistrust or confusion at the point of consent.

Feasibility. Moderate. Enforcement relies on SOTTOs which in many cases are staffed by officers holding the role as an additional charge (NAMS, 2024).

4.3 Option Three: Implement a family navigation and referral system at the district-level

Description. This option creates a district structure that links every ICU and trauma hospital in the district to the donation network. It combines identification protocols with a 24-hour one-call referral line, a shared pool of rapid-response coordinators employed by the SOTTO, multilingual family-navigation materials, and stage-by-stage reporting reviewed by a District Organ Donation Committee.

Benefits. It addresses institutional and family barriers together, and sharing coordinators across a district closes the gap Option Two leaves for hospitals that cannot justify a full-time post. Indian evidence suggests trained counselling works. Jha et al. (2025) found that families of 55.7 percent of 158 medically suitable brain-stem dead patients consented after counselling by a dedicated coordinator team at a government tertiary hospital in 2009–2021. Among families of medico-legal autopsy cases at AIIMS, New Delhi, about 31 percent consented when approached by a doctor with a social worker, against about 12 percent when approached by a social worker alone (Kumar et al., 2025). In Tamil Nadu, medical colleges and district hospitals with retrieval-centre status produced 16 percent of donors in 2024 (Natarajan & Ethirajan, 2026). The reporting component would also make losses visible.

Costs. The ten-district pilot is estimated at INR 2.3–3.3 crore a year, using NOTP unit rates with a contingency for training and materials (Table 6, section 6.3), about 11–16 percent of the NOTP's proposed 2025–26 outlay (NAMS, 2024). Costs rise with the number of districts, not hospitals: three coordinators serve all hospitals in a district, which is cheaper per hospital than giving each hospital its own coordinator.

Risks. It needs to be coordinated between health departments, SOTTOs, hospitals and police and sustained funding beyond a pilot. If district roles are not protected, district coordinators may be absorbed into other duties when there are few cases to coordinate. The impact of its effect has not been tested in India yet as mentioned in section 5.4.

Feasibility. Moderate to high if added as a funded NOTP component. It is linked with existing structures such as SOTTOs, e-Pratyaropan platform (NAMS, 2024; PIB, 2026), and gives effect to the designated nodal officers and audits of missed opportunities called for in the 2026 guidelines (NOTTO, 2026b, as reported in ETV Bharat, 2026).

4.4 Reforms that were under discussion, but not selected

Presumed consent (opt-out). Spain has an opt-out law, but families still decide, and analysts credit its coordinator network rather than the law (The Lancet, 2024); Ireland's experience points the same way (Athavle & Gopal, 2026). In India, where fear of organ trafficking already shapes family decisions, presuming consent could deepen mistrust.

Donation after circulatory death (DCD). Spain is a world leader in donation after the heart has stopped (La Moncloa, 2025), but in India such donations remain in single digits and lack national guidelines

(Athavle & Gopal, 2026). DCD needs separate protocols and safeguards, so it is recommended as a parallel reform.

5. RECOMMENDED POLICY OPTION

This paper recommends Option Three, incorporating the mandatory clinical triggers from Option Two. Without support, mandatory identification may produce paper-based compliance; without mandatory identification, support may be bypassed. They are combined in a single intervention, with hospitals identifying and referring and the district system ensuring a timely, competent response.

Table 4. Comparison of policy options

Criterion	**Option One: Awareness**	**Option Two: Mandates**	**Option Three: District system**
Addresses in-hospital identification	No	Yes	Yes
Addresses family trust	Partly	No	Yes
Supports hospitals without coordinators	No	No	Yes
Generates pathway data	No	Partly	Yes
Indicative cost	INR 1.2 crore a year	INR 4.2 lakh per hospital a year	INR 2.3–3.3 crore a year (pilot)
Likely effect on donations (judgement)	Low	Moderate, limited by weak enforcement	Potentially high, but untested in India

Note: Costs use NOTP budget lines and unit rates (NAMS, 2024; Government of India, 2021). Other ratings are the author's qualitative judgement based on the evidence in section 3, not measured estimates; see section 5.4.

5.1 Justification

- It focuses on the binding constraint. The analysis shows that losses occur at identification, referral and family approach, which is exactly where the framework intervenes.
- It builds upon the best practices in India. Tamil Nadu's results came from hospital readiness, simple registration for retrieval centres, standard protocols and trained counsellors, all of which the framework adapts to the district level.
- It puts current policy into action. It needs no new legislation and gives practical effect to the 2011 THOTA amendment, the 2024 reporting directive and the 2026 brain-stem death guidelines.
- The direction of effect has independent support. Consent reached 55.7 percent where dedicated coordinators counsel families of brain-stem dead patients (Jha et al., 2025), and was higher when doctors joined counselling in a separate autopsy-based study (Kumar et al., 2025). Although these settings are not directly comparable, they point in the same direction as the gains at Tamil Nadu's

retrieval centres and Spain's coordinator model. The size of the effect in Indian districts has yet to be measured.

- It has low regret and is testable. A pilot with comparison districts allows rigorous evaluation, and even a modest effect would give India the stage-by-stage data it currently lacks.

5.2 Expected benefits

The goal of the framework is to increase referral rates, approach and consent rates, expand donation to districts without donors, and alleviate pressure on living donors.

5.3 Challenges and mitigation

Challenge	Mitigation
Limited ICU and retrieval capacity in some districts	Select pilot districts with at least one ICU-equipped hospital; link them to a hub transplant centre; simplify NTORC registration
Clinician fear of medico-legal consequences	Issue a central clarification on brain-stem death certification; train police; adopt Tamil Nadu's practice of allowing the treating hospital's local police station to begin medico-legal procedures
Coordinator shortages and burnout	Create a district pool of rapid-response coordinators; fund posts through NOTP; offer certification and career progression
Family mistrust about allocation and trafficking	Explain allocation rules and safeguards in navigation materials; include independent doctors on certification panels; publish district-level data
Unreimbursed donor maintenance costs	Remove the five-donor cap and government-hospital condition on maintenance support; reimburse all certified donor maintenance
Low participation by private hospitals	Link brain-stem death reporting and NTORC registration to licence renewal and insurance empanelment; reimburse maintenance; publish hospital-level data
Inter-agency coordination	Constitute a District Organ Donation Committee chaired by the District Collector or Chief Medical Officer, with defined roles and monthly reviews

5.4 Limitations of this analysis

There are three limitations. First, the ratings in Table 4 and the benefits above are qualitative judgements, not measured estimates. They draw mainly on Tamil Nadu, Kerala and Spain, and no study has yet tested

a district referral and family navigation system in India. Tamil Nadu built its programme over more than 15 years, with dense private hospital networks and strong NGO support (Abraham et al., 2012; Natarajan & Ethirajan, 2026), so the same design may yield smaller gains in states with fewer ICUs and less administrative capacity. This uncertainty is the main reason the paper recommends a pilot rather than national rollout.

Second, the evidence has gaps. Because India does not count possible donors, referrals, family approaches or refusals, neither the losses at each stage of Figure 1 nor the framework's likely effect can be estimated. Much national data comes from official summaries and their media reporting, and the state analysis in section 3.2 has the limits noted there.

Third, the framework does not resolve several other barriers. It does not fix the inconsistent legal definition of death in the Registration of Births and Deaths Act and the Bharatiya Nyaya Sanhita, which leaves some clinicians fearful of police inquiries (Athavle & Gopal, 2026). It does not develop donation after circulatory death (section 4.4). It cannot create ICU or retrieval capacity in districts with none, and it does not address the cost of transplantation and post-transplant medicines, which keeps access skewed towards patients who can use private hospitals (NAMS, 2024; Kute et al., 2026). These need national reforms.

6. IMPLEMENTATION FRAMEWORK

6.1 The five components Component 1: Donor Identification Triggers. A one-page checklist is used by all ICUs and trauma units. A ventilated patient with a catastrophic brain injury and a severely reduced Glasgow Coma Scale score, at a threshold set in line with the 2026 national guidelines, triggers a referral to the district coordinator, irrespective of any discussion with the family about donation. A referral is not a donation; it ensures that the case is assessed and that the family is later offered the choice. Determination of death is kept strictly separate from donation: certification is done by the authorised board, transplant surgeons and coordinators have no role in this process and donation is approached only after death has been certified, as per the requirement of the THOTA certification and also international practice (Government of India, 1994, 2014).

Component 2: One-call referral system. Every district has a single, verified 24-hour referral number linked to a district nodal coordinator and the SOTTO, displayed in every ICU.

Component 3: Rapid-response transplant coordinators. A district pool of trained coordinators, employed by the SOTTO or the district health society, visits or supports by tele-consultation any hospital that makes a referral, helping with certification logistics, donor maintenance, family counselling and communication with the police.

Component 4: Multilingual family-navigation materials. A short guide written in the local language and illustrated to help families understand brain-stem death, the consent process, how organs are allocated, that there is no cost to the family, how the body is returned for the last rites and the safeguards against organ trading. These materials are for informing, not persuading: coordinators should counsel without pressure, and a family's decision to decline must be respected and recorded as a legitimate outcome, not treated as a failure of information (Natarajan & Ethirajan, 2026).

Component 5: Implementation-focused reporting. Hospitals report monthly to the SOTTO on ICU deaths, possible brain-stem death cases, referrals, certifications, family approaches, consents, retrievals and reasons for non-donation. These data feed a district dashboard reviewed by the District Organ Donation

Committee. Crucially, the data flow back: each hospital receives a quarterly report showing how many potential donors it identified, how many were referred and certified, and where the others stopped, so it can fix its own problems. This mirrors the NAMS Task Force's call for monthly reviews of ICU deaths for missed opportunities (NAMS, 2024) and Tamil Nadu's quality-improvement review of every donation (Natarajan & Ethirajan, 2026).

Cross-cutting requirement: training before rollout. Having a protocol on paper is not enough to change practice unless staff are aware of the reason for the trigger, what it does and who they need to contact when triggered. Before launch, certification panels and coordinators in each pilot district would receive the structured, simulation-based training recommended in the 2026 guidelines, and each team involved in the emergency response would be given a short-term, practical training on the checklist and referral pathway (NOTTO, 2026b, as reported in ETV Bharat, 2026). This addresses the training and workload deficits that transplant coordinators report (Vincent et al., 2024).

6.2 Responsibilities

Table 5. Roles and responsibilities

Institution	Responsibilities
MoHFW / NOTTO	Fund through the NOTP; issue protocols and materials; set indicators; commission evaluation
ROTTOs and SOTTOs	Host referral lines; employ coordinators; compile data
State health departments	Select districts; simplify NTORC registration; agree police protocols
District committee	Chaired by the District Collector or CMO; monitor compliance monthly
Hospitals with ICUs, public and private	Apply triggers; refer; provide counselling space; report data
Medical colleges and NGOs	Train staff; supply retrieval teams; support translation and families

6.3 Resources

Table 6 uses NOTP unit rates (Government of India, 2021; NAMS, 2024) for a pilot of 10 districts and 3 coordinators per district. Because donor numbers are uncertain, it shows a low scenario of 25 donors a year and a high scenario of 100, which correspond roughly to the 2024 national rate (0.81 per million) and Tamil Nadu's rate (3.48 per million) applied to a pilot population of 25 to 30 million. Evaluation is costed at the NOTP's own 2025–26 evaluation allocation, and items with no fixed NOTP rate, such as training, translation, the reporting dashboard, and the extension of the helpline, are allocated a 25 percent contingency, which will be replaced with detailed costing in Phase 0.

This equals about 11 to 16 percent of the NOTP's proposed 2025–26 outlay of INR 20.83 crore, or INR 4.6 to 6.7 crore over 24 months (NAMS, 2024). Part could be met from under-used NOTP lines, with state co-financing.

Transplantation is also the most cost-effective long-term treatment for kidney failure: a 2025 systematic review found it the most cost-effective option in 9 of 13 cost-utility studies in high- and middle-income countries (Wulandari et al., 2025). An Indian study of 493 patients estimated haemodialysis at INR 5.5 lakh a year against a one-off transplant cost of about INR 8.8 lakh, so costs equalise within about 20 months (Trivedi & Sodani, 2024). Savings vary by setting, since dialysis is much cheaper in many public facilities and transplant recipients need lifelong medicines. At the higher estimate, 25 additional donors would save about INR 2.75 crore a year in dialysis costs, close to the pilot's annual cost; savings would be smaller in the public sector. The main case therefore rests on survival and quality of life, with fiscal savings a secondary benefit.

Table 6. Indicative annual cost of the ten-district pilot

Cost item	Unit rate or basis	Low scenario (INR)	High scenario (INR)
District coordinators (30)	INR 35,000 per month (NOTP)	1.26 crore	1.26 crore
Donor maintenance and funeral support	INR 1.1 lakh per donor (NOTP)	27.5 lakh (25 donors)	1.10 crore (100 donors)
Independent evaluation	NOTP evaluation line, 2025–26	30 lakh	30 lakh
Training, materials, dashboard, helpline	25% contingency (assumption)	46 lakh	67 lakh
Total per year		about 2.3 crore	about 3.3 crore

Source: Author's calculation using NOTP unit rates and budget lines (Government of India, 2021; NAMS, 2024); see Supplementary Table S1.

6.4 Centre–state roles

Public health and hospitals fall under Entry 6 of the State List, and THOTA was enacted under Article 252(1), so it applies in a State only if that State adopts it by resolution (Government of India, 1994). In 2025 the Supreme Court suggested that the Union, through NOTTO and in consultation with the States, evolve a five-year plan to equip hospitals in every State (Indian Society of Organ Transplantation v. Union of India, 2025). The framework should therefore be voluntary for States and attractive to them. The Centre would fully fund the pilot through the NOTP and set common protocols and indicators through NOTTO; States will choose the districts, house coordinators in their SOTTOs and agree police procedures with their home departments. At scale, funding would follow the centrally sponsored scheme model, with performance grants tied to referral and consent indicators. There is also a fiscal incentive for States to participate because for every kidney transplant, the long-term costs of dialysis are reduced under State and national insurance systems.

Private hospitals must be brought in on equal terms, since about 87 percent of kidney transplant centres are private (Kute et al., 2026). Private ICUs would follow the same triggers and reporting, with brain-stem death reporting and NTORC registration made conditions of transplant licence renewal and of empanelment under public insurance schemes, and with donor maintenance reimbursed so that referral carries no financial penalty.

6.5 Timeline and accountability

Table 7. Phased implementation plan

Phase	Period	Key activities
0: Design	Months 0–6	Finalise protocols and indicators; select five states and ten districts plus comparison districts; train coordinators; collect baseline data
1: Pilot	Months 6–30	Launch triggers and referral lines; deploy coordinators; begin reporting and quarterly reviews
2: Evaluation	Months 24–36	Independent evaluation; revise and publish
3: Scale-up	Years 3–5	Phased national expansion, prioritising low-donation states

Pilot states should mix one high performer (Tamil Nadu or Telangana) with low-donation states such as Uttar Pradesh, Madhya Pradesh, Rajasthan and Odisha. Hospitals answer to district committees, districts to SOTTOs, and NOTTO publishes annual district results; compliance should be linked to registration renewal.

7. EXPECTED OUTCOMES

Short term (0-12 months): All pilot ICUs begin to use the checklist and the referral number, monthly reporting becomes routine and the first district baselines of possible donors are available.

Medium term (1-3 years): most identified cases are referred and most families are approached by trained staff, consent rates rise, and districts that previously had no donations begin to record donors.

Long term (3-7 years): the national rate may increase beyond 1 per million, the reliance on living donors is reduced and published data will help build public confidence.

8. MONITORING AND EVALUATION

Table 8. Monitoring framework

Stage	Indicator	Data source
Input	Coordinators in post; staff trained	SOTTO records
Identification	Possible brain-stem death cases per 100 ICU deaths	Hospital reports; death audits
Referral	Share of possible cases referred on time	Referral line logs
Certification	Share of referred cases certified	Form 10; SOTTO data
Family approach and consent	Share of families approached by trained staff; consent rate; refusal reasons	Coordinator records
Outcome	Donors; organs transplanted; donors per million by district	NOTTO registries

Hospitals report monthly, districts quarterly, and NOTTO publishes annual results, with a mid-term review at 12 months. An independent institution would evaluate the pilot, using a difference-in-differences design that compares each pilot district with a comparison district matched on population, ICU beds, presence of a transplant centre, road-traffic deaths and baseline donation. Because a district may record only a few donations a year, completed donations are too rare to be the main outcome within 24 months. The primary outcomes are the proportion of possible brain-stem death cases referred and the proportion of certified cases whose families are approached by trained staff. Success is defined in advance as at least 80 percent on both measures and a statistically significant gain over comparison districts; the 80 percent stretch target will be checked against Phase 0 baseline data. Donation rates are a secondary outcome, pooled across the ten districts and tracked into Phase 3. Retrospective ICU death audits, as used in Spain, would estimate missed donors (Matesanz et al., 2017).

9. CONCLUSION

India does not lack organ donation law, institutions or public goodwill; it lacks a reliable way of turning them into donations at the bedside. Brain-stem death goes unrecognised in hospitals, smaller hospitals do not know whom to call, families are approached without trained support, and no one measures the stages in between. The contrast between Tamil Nadu and states with almost no deceased donation, under the same law, shows these losses are preventable.

A District Organ Donation Referral and Family Navigation Framework offers a practical, testable way to close this last mile. Piloted carefully and scaled in phases, it could move India beyond pledges to a system in which every eligible family has an informed choice, and that choice is respected.

POLICY RECOMMENDATIONS

Recommendation 1: Implement a standard donor identification and referral checklist at all ICU/trauma hospitals in accordance with the national guidelines (2026).

Recommendation 2: Institute district nodal coordinators and a 24-hour referral line with SOTTO, with rapid response coordinators.

Recommendation 3: Create family navigation materials in multiple languages and offer a confidential counselling room in each of the referring hospitals.

Recommendation 4: Train staff in the ICU, emergency and district hospitals, and the police in identification, certification and referral and simplify the registration of NTORC.

Recommendation 5: Disseminate indicators at stages, provide indicators as feedback to hospitals and fully reimburse hospitals for the costs of maintaining donors.

Recommendation 6: Judge awareness campaigns by family conversations, referrals and completed donations, not by pledges alone.

DECLARATIONS

Use of Artificial Intelligence. The author declares no use of artificial intelligence through the course of this paper. Ethics and consent. This paper uses only published data and literature along with one illustrative interview conducted by the researcher for social, behavioural and educational purposes involving one human participant working as a transplant coordinator in a tertiary care hospital having a transplant programme. The protocol was determined to fall under Exempt Category 2 (Surveys/Interviews) involving minimal risk. Informed voluntary verbal consent was obtained from the participating transplant coordinator prior to the interview. No personal health records or identifiable patient data were collected. The participant was informed that the anonymised transcript of the operational workflow dialogue would be included as an appendix in this manuscript. To preserve strict confidentiality, all explicit institutional names, specific geographical locations, and individual professional identities have been thoroughly redacted. The participant is referred to generically as "Transplant Coordinator at a Tertiary Care Facility." Conflicts of interest. The author declares no conflicts of interest. Funding. This research received no specific funding. Data availability. All data used are publicly available from the sources cited. The calculations behind Tables 3 and 6 and Figure 2 are provided in Supplementary Table S1.

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APPENDIX A. ILLUSTRATIVE INTERVIEW

Q1. So what do you do and what type of hospital do you work in?

I am a transplant coordinator in a tertiary care hospital having a transplant programme. My job is to ensure effective communication between the ICU staff, the transplant physicians and the family and the appropriate organ-donation organisations. Much of the job is communicating with families and making sure that the various phases of the donation process are completed in the right order.

Q2. How many cases do you see that have the potential to result in brain-stem-death each month and how many reach family counselling?

This fluctuates significantly between each month. Not all patients with severe neurological injury will go on to fulfil the criteria for brain-stem-death certification. Some become medically unsuitable, and others do not reach formal certification. Above all, the potential donor has to be identified, and identification can be inconsistent.

Q3. Could you walk me through a typical case from the ICU to consent? Where does the process most often break down?

The first time that the ICU team becomes aware that irreversible neurological injury has been sustained, it is typically when a patient's condition is deteriorating. The necessary clinical tests are then conducted and, if necessary, brain-stem-death certification is issued when the requirements are fulfilled. The family then receives counselling after certification about organ donation.

There isn't necessarily one point where the system breaks down. There can be delays in the identification, certification process, process between teams, or in family counselling. More often, a few minor logistical issues combine and have an impact on the process.

Q4. How are potential donors identified? Is there a checklist, and who usually notices?

A treating medical team or the ICU will usually have the best idea of who to consider as a donor since they are watching the patient's neurological status. A standardised trigger or checklist can help to ensure that donation is a consistent process as opposed to entirely dependent on the clinician's memory to start it.

However, the checklist needs to be accompanied by training. Staff should be familiar with the meaning of the trigger and who to contact following that.

Q5. Are there ever any referrals made to you at smaller or non-transplant centres? In what ways do you hear about them?

They can, because not all potential donors will show up at a hospital where they can have a transplant. The difficulty is frequently the person you should speak to and the information you want to share.

It is helpful to have a clear referral process, especially if hospitals are not used to dealing with organ donation. Referrals can be more efficiently coordinated if the contact process is simple and is open at all hours.

Q6. Are there difficulties in accessing certification (e.g. police/legal issues, apnoea test, availability of panels)?

Brain-stem-death certification must meet certain clinical and legal criteria, and thus several people and processes will be required to be coordinated. Sometimes requirements for the necessary specialists can influence timing. It can also be difficult to undertake the required evaluations in a critically ill person.

At times a second level of coordination is provided by police or medico-legal processes. I would describe these as several logistical barriers rather than one single obstacle.

Q7. How and when are families engaged, and in what languages? What written materials do you use?

Timing in the conversation is very important. Before discussing brain-stem death with the family, the patient's medical condition and what it means should be understood.

Also, language is important. Families should be counselled in a language they are comfortable in. While written information may be helpful, it cannot be a substitute for a conversation as families may have questions that are unique to them.

Q8. What are the most common reasons families refuse? What helps them decide?

There isn't one universal reason. Families can have religious beliefs or cultural issues, may lack a full understanding of brain-stem death conditions or may simply be overwhelmed. Many also worry about the respect and treatment of their relative and about customary funeral practices as well.

Communication is crucial and it is best to be clear and concise. Families should have the time and information necessary to make a meaningful decision without being rushed to make a donation.

Q9. In reality, does it matter if the patient had already made a commitment to donate?

Can help make the conversation easier, as the family may already be aware that the patient wanted to donate. It can give them a sense that they are respecting the patient's wishes.

But a pledge doesn't eliminate the need for counselling. Questions remain within families, and of course the patient's medical situation and the procedures involved determine the eventual outcome of the donation.

Q10. Who pays for keeping the donor on support, and does the cost deter hospitals?

Continued care in the ICU is required, as well as ongoing monitoring, investigations, and staff time, for maintaining a potential donor. Financial arrangements may differ from hospital to hospital.

Resource limitations may apply, especially if there are not enough beds in the ICU. This is another key benefit to coordination and efficient identification. But do not overlook the non-cost factors, such as clinical, logistic and institutional considerations.

Q11. Does NOTTO's monthly brain-stem death report get filled? What does it miss and what does it capture?

Reporting is useful because it provides us with a little bit of an idea of how many potential donors are being identified. However, without a reason next to the number, that number alone does not provide any explanation as to why that case did not lead to donation.

For instance, the report does not always differentiate between medical unsuitability, certification issues, family refusal and logistical issues being the reasons for the case being stopped. It is important to know where cases are being lost along the pathway when working on improving the system.

Q12. These are my 5 framework components. What would make the idea better, what would not work, and what would be lacking?

These five components make sense because donation is really a path, not one decision. Identification and referral are especially crucial, because a family cannot be approached about donation if the potential donor is never identified or referred.

Also, I would add some kind of feedback system. Hospitals should be able to see the number of potential donors they identify and where cases are 'stuck'. It would also be essential for training and the clear definition of responsibilities, since a framework cannot be in place merely on paper.

Q13. What would be one thing you would change if you had the chance?

I would work on standardising the process. All ICUs should be aware of who identifies a potential donor, who to contact as soon as a potential donor is identified, what happens after referral and who will be responsible for communicating with the family.

The process should be family-centred at the same time. The aim must not only be to get more donors; it must also be to identify potential donors properly and to provide families with correct information and allow them to make informed decisions free from pressure.

APPENDIX B. SUPPLEMENTARY TABLE S1

This appendix presents the calculations performed to prepare Tables 3 and 6 and the figures given in section 3.2.

Method. Deceased donors per million population (pmp) = Deceased donors in the year / (Projected population (in thousands) / 1000). The NOTTO donation figures (2025b) are NOTTO figures based on the state where donations were made, and are cross-checked for the five leading states with the Kute et al. (2026) figures. Populations are projections for 1 March of each year (National Commission on Population, 2019).

Table S1a. Inputs and donor rates by state, 2023 and 2024

State	Donors 2023	Population 2023 ('000)	pmp 2023	Donors 2024	Population 2024 ('000)	pmp 2024
Telangana	252	38,090	6.62	188	38,272	4.91
NCT of Delhi	66	21,359	3.09	n/a	21,752	n/a
Karnataka	178	67,692	2.63	162	68,115	2.38
Tamil Nadu	178	76,860	2.32	268	77,089	3.48
Gujarat	146	71,507	2.04	119	72,367	1.64
Maharashtra	148	126,385	1.17	172	127,360	1.35
Andhra Pradesh	41	53,156	0.77	n/a	53,340	n/a
West Bengal	16	99,084	0.16	n/a	99,563	n/a
Chhattisgarh	4	30,180	0.13	n/a	30,524	n/a
Madhya Pradesh	8	86,579	0.09	n/a	87,610	n/a
Rajasthan	7	81,025	0.09	n/a	81,897	n/a
Odisha	3	44,291	0.07	n/a	44,420	n/a
Uttar Pradesh	3	235,687	0.01	n/a	238,078	n/a
India	1,099	1,386,180	0.79	1,128	1,398,598	0.81

Sources: NOTTO (2025b); Kute et al. (2026); National Commission on Population (2019). n/a = not separately available.

Table S1b. Summary calculations reported in section 3.2

Calculation	Value
Five leading states, 2023: share of India's population	27.5%
Five leading states, 2023: share of deceased donors (902 of 1,099)	82.1%
Five leading states, 2024: share of population/share of donors (909 of 1,128)	27.4% / 80.6%
Six large states, 2023: population ('000) and share of India	576,846 (41.6%)
Six large states, 2023: donors, share of donors and combined pmp	41 (3.7%); 0.07 pmp
Sensitivity: six states' combined pmp if true donors were five times higher	0.36 pmp
Ratio of Telangana to Uttar Pradesh pmp, 2023	about 520
Six states' donors at Tamil Nadu's 2023 rate (2.32 pmp)	about 1,336
All India at Tamil Nadu's 2024 rate (3.48 pmp): implied donors, 2024	about 4,862 (actual 1,128)
Organs transplanted per deceased donor, 2024 (3,403 ÷ 1,128)	3.02
Implied additional transplants per year	about 11,266

Source: Author's calculations from Table S1a and NOTTO (2025a, 2025b). Six large states: Uttar Pradesh, West Bengal, Madhya Pradesh, Rajasthan, Odisha and Chhattisgarh.

Table S1c. Inputs and calculations for the pilot cost estimate (Table 6)

Item	Low scenario (INR)	High scenario (INR)
Coordinators: 30 × INR 35,000 × 12 months	1,26,00,000	1,26,00,000
Donor maintenance and funeral support: INR 1.1 lakh per donor (25 / 100 donors)	27,50,000	1,10,00,000
Independent evaluation (NOTP evaluation line, 2025–26)	30,00,000	30,00,000
Subtotal	1,83,50,000	2,66,00,000
Contingency at 25% (training, materials, dashboard, helpline)	45,87,500	66,50,000
Total per year	2,29,37,500	3,32,50,000
Share of NOTP proposed outlay 2025–26 (INR 20.83 crore)	11.0%	16.0%
Total over 24 months	4,58,75,000	6,65,00,000

Source: Author's calculations using NOTP unit rates and budget lines (Government of India, 2021; NAMS, 2024). The 25 percent contingency is an assumption.

Limitations. Donors are counted where donation takes place and not by their place of residence, which can artificially inflate the rates in the referral centres like Delhi. Population estimates are based on the 2011 Census. Where NOTTO reports state-level 2024 figures, these are used; otherwise, the 2024 figures are not reported. Rates published elsewhere are not directly comparable, as they are based on different years or population estimates.